

Artificial Evolution Status Report

Current problem summary, root-cause findings, and latest experiment outcomes as of 2026-05-10

Experiment Timeline

Exp	Purpose	Result	Tick	Births	Matured	Key note
36	first-tool full test	too fast	1-2	-	-	first tool appeared almost imm
39	emergent technology	emerged early	8	9	6	proto_composite_type_a at tick
40	first sexed world	extinct	201	26	18	failed before gen3
41	full body search	no survivor	385	42	31	best body still extinct
42-44	diagnostics	root cause found	-	-	-	gen1 blocked by low energy an
45	household continuity fix	success	275	34	30	reached gen3 with same hard

English Section

1. Current Problem Set

- The project has moved away from explicit named crafting and toward emergent technology defined by behavioral impact and spread.
- The old named-tool setup was too easy: the first tool appeared at tick 1-2, which made the research question too weak.
- The new emergent-technology setup is conceptually better, but technology still appears too early: ``proto_composite_type_a`` emerged at tick 8 in the smoke test.
- Under sexed reproduction, the main failure mode before the fix was a broken generation-1 to generation-2 reproduction chain.

2. Major Experiment Outcomes

- Experiment 36 first-tool study: all runs succeeded, but too quickly, with mean first-tool tick 1.5.
- Experiment 39 emergent technology study: `proto_composite_type_a` emerged at tick 8 with peak population 21.
- Experiment 40 first sexed-generation test: `final_tick=201`, `births=26`, `matured_children=18`, `extinct=True`.
- Experiment 41 full sexed-world body search: 250 candidates across 3 seeds each; best body was `idx=8` but still went extinct at tick 385.
- Experiment 45 after the family-household fix: `final_tick=275`, `extinct=False`, `gen3=True`, `gen3_tick=275`.

3. Root Cause Found

- Generation-1 agents did have valid sex assignments and inherited parent metadata.
- Diagnostics in experiments 42-44 showed repeated ``repro_fail`` cases where adult generation-1 females had mates but hit ``nest_food=0`` and ``low_energy`` together.
- This pointed to a household continuity problem: children matured without reliably resolving usable family household storage.
- The fix therefore targeted family household ownership and storage continuity rather than making the world easier.

4. What Is Solved vs Still Open

- Solved: generation-3 adulthood can now be reached in the same hard world with the same body family.
- Solved: research-grade telemetry, readable logs, and dashboard outputs exist for serious analysis.
- Open: the full sexed-world search must be rerun after the fix because the old ranking is now stale.

- Open: emergent technology still appears too early, so the world pressure and experimentation loop need to be hardened.
- Open: publication-grade inference still needs multi-seed batches, even though the data protocol is already defined in C:\artificial-evolution\docs\PAPER_DATA_PROTOCOL.md.

Key Reference Files

- C:\artificial-evolution\data\first_technology_emergence_study.md
- C:\artificial-evolution\data\sexed_world_body_search.md
- C:\artificial-evolution\data\sexed_generation_test.md
- C:\artificial-evolution\data\experiments\experiment_040.md
- C:\artificial-evolution\data\experiments\experiment_041.md
- C:\artificial-evolution\data\experiments\experiment_044.md
- C:\artificial-evolution\data\experiments\experiment_045.md
- C:\artificial-evolution\data\dashboards\sexed_generation_mode\experiment_045_sexed_seed_7\index.html
- C:\artificial-evolution\data\research_runs\experiment_033_paper_capture\manifest.json
- C:\artificial-evolution\docs\PAPER_DATA_PROTOCOL.md